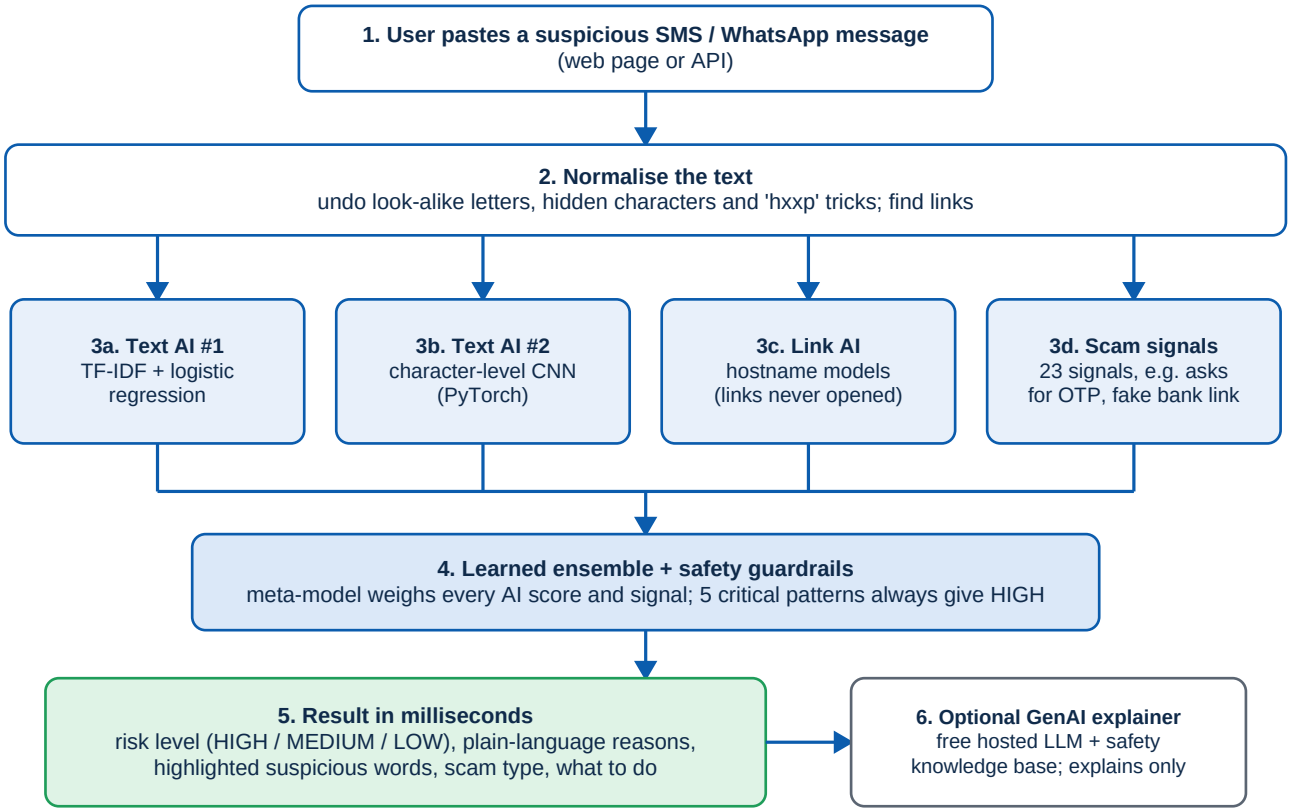
ScamShield

AI that tells you, in plain words, whether a text message is a scam | IBM Z Datathon 2026 (AI for Good)

Abstract

SMS and WhatsApp scams (fake KYC alerts, "your parcel is held" fees, UPI refund tricks and "digital arrest" threats) hit elderly people and first-time smartphone users hardest, and once an OTP is shared the money is usually gone. Spam filters say "spam" without saying why, and online link checkers send private messages to third parties. **ScamShield** is an AI system that analyses a pasted message and returns, in milliseconds, a risk level (HIGH / MEDIUM / LOW), reasons written for non-technical readers, the exact words that looked suspicious, and what to do next. It combines two text-classification models (a TF-IDF model and a small PyTorch character-level neural network), link-reputation models that judge a web address without ever opening it, and 23 interpretable signals, fused by a learned ensemble with hard safety guardrails. A generative-AI layer can add a plain-language explanation grounded in a curated safety knowledge base, but it never changes the verdict. All models are trained, evaluated and served on **IBM LinuxONE (s390x)**, where we also measure latency and throughput.

How it works: end-to-end flow



Everything above runs on IBM LinuxONE (s390x); the LLM is the only external call.

The ensemble is a logistic-regression meta-model trained on held-out data, so it learns how much to trust each model and each signal. Thresholds are chosen on out-of-fold scores, and a MEDIUM ("be careful") band is always kept.

The problem

- Victims are mostly elderly and low-digital-literacy users who lose money within minutes of sharing an OTP or approving a UPI request.
- Generic spam filters give no explanation, so users cannot learn to recognise scams, and false alarms on genuine bank messages destroy trust.
- Link scanners usually upload the message or link to an outside service, which leaks private conversations.

What we are doing

- **Detect:** classify a message with several AI models and a learned ensemble, trained on public SMS and phishing datasets plus Indian-context messages we wrote ourselves.

- **Explain:** show reasons in plain English, highlight the words that drove the score, and name the scam type (KYC, parcel, job, electricity, impersonation, UPI refund, and more).
- **Protect:** never store the message, never open links, and always flag critical patterns such as "share your OTP" or a look-alike bank website.
- **Prove it on IBM Z:** run the whole pipeline (data, training, evaluation, serving, benchmark) on the LinuxONE server and keep the evidence.

AI components

Component	What it does	Why it is there
TF-IDF + logistic regression	Scores the wording of the message using word and character n-grams.	Fast and strong on short texts; its weights show which words matter.
Character-level CNN (PyTorch, about 100k parameters)	Reads raw characters with convolutions; trained from scratch.	Robust to tricks like "0TP" and look-alike letters; no pretrained download; trains in minutes on 3 CPUs.
Link AI	Gradient boosting on 18 lexical features plus a character n-gram model, averaged; official brand domains are allow-listed.	Catches fake bank and delivery domains without visiting them.
Scam signals and guardrails	23 signals: 20 warning detectors (OTP request, brand look-alike, digital arrest, UPI PIN to receive money, and more) and 3 genuine-pattern signals.	Give the ensemble extra features, produce the user-facing reasons, and force HIGH for 5 critical patterns.
GenAI explainer (optional)	Free hosted LLM with retrieval over a safety knowledge base; output is validated, with a built-in fallback.	Friendlier wording; it explains the decision but cannot change it.

Role of IBM LinuxONE

IBM Z hosts the banking and telecom systems where this kind of scanning would be deployed: a provider can score very large message volumes next to its own systems without sending sensitive content to outside clouds. In this project the LinuxONE server (s390x, 3 vCPU, 8 GB) is where every model is trained, where evaluation and tests run, and where the API and web page are served. Training on the server also avoids byte-order problems: s390x is big-endian, so model files from a laptop could be misread; ScamShield records the training architecture and refuses to load a mismatched model. Every number in this document was measured there.

Results (measured on IBM LinuxONE)

Test set (measured on IBM LinuxONE, s390x)	PR-AUC	Scams caught (HIGH)	False alarms (HIGH)
Unseen public SMS data (916 messages)	0.987	96.9%	0.5%
Unseen Indian-context templates (461 messages)	0.997	100%	6.6%

All models were trained on the server (IBM z17 VM) in about 11 minutes, and all 48 tests pass there. On public SMS data the full system (0.987) is on par with a plain TF-IDF model (0.988); its gain shows on the Indian-context set (0.997 vs 0.967). Genuine OTP and bank alerts rated HIGH: 5.3% (7 of 132) on this split, and 1.2% on average (range 0-2.3%) across five different template splits. Link AI on 33,373 unseen domains: PR-AUC 0.920. Median analysis time 2.7 ms per message (p95 8.5 ms), about 250 messages/s on one thread. The 36 adversarial test messages (34 passed) were written by the team, so they are a regression check, not independent proof.

Responsible AI, privacy and limits

- Messages are never stored or logged, links are never visited, and any LLM sees only scrubbed text (links, phone numbers, amounts and numbers replaced).
- ScamShield is advisory only: it never blocks anything, never says "safe" (LOW means "no common scam signs"), and every result carries a disclaimer.
- Known limits: English only; public training data is dated; Indian coverage relies on our own templates; genuine messages in unfamiliar styles (for example a housing-society notice or a UPI payment receipt) can be over-flagged, and novel scam scripts without warning signs can be missed.

Technology and future work

Stack: Python, scikit-learn, PyTorch, Flask, plain HTML/JavaScript; datasets: UCI SMS Spam Collection, Mendeley SMS Phishing Dataset, PhiUSIIL Phishing URL Dataset (all CC BY 4.0). **Next:** regional-language scripts, screenshot input, a WhatsApp/Telegram bot, and a provider-side batch API benchmarked on LinuxONE.